

## REACTION MECHANISM

- Polarisation of  $\sigma$ -bond caused by polarisation of adjacent  $\sigma$ -bond is referred as—  
 (1) Mesomeric effect      (2) Electromeric effect  
 (3) Inductive effect      (4) Hyperconjugation
- Select the molecule with more polar bond from each of the indicated bonds among given set of molecules:  
 (a)  $\text{H}_3\text{C} - \text{H}$ ,  $\text{H}_3\text{C} - \text{Br}$   
           (I)                (II)  
 (b)  $\text{H}_3\text{C} - \text{NH}_2$ ,  $\text{H}_3\text{C} - \text{OH}$   
           (I)                (II)  
 (c)  $\text{H}_3\text{C} - \text{OH}$ ,  $\text{H}_3\text{C} - \text{SH}$   
           (I)                (II)  
 (1) I, I, I                                (2) II, II, II  
 (3) II, II, I                              (4) II, I, II
- Which of the following is a correct statement:  
 (1) Canonical structures are real structures and explain all the properties of a real molecule  
 (2) Inductive effect increases as the number of intervening bonds increases  
 (3) Hyperconjugation is a temporary effect  
 (4) Canonical structures, contribute to the actual structure in proportion to their stability.
- Which of the following does not represent a correct set of nucleophile—  
 (1)  $\text{HS}^\ominus$ ,  $\text{C}_2\text{H}_5\text{O}^\ominus$   
 (2)  $(\text{CH}_3)_3\text{N}$ ,  $\text{NH}_2^\ominus$ ,  $\text{CN}^\ominus$ ,  
 (3)  $\text{SO}_3$ ,  $\text{H}_2\text{O}$ ,  $\text{CH}_3^\ominus$   
 (4)  $\text{CH}_3 - \text{S} - \text{CH}_3$ ,  $\text{NO}_2^\ominus$ ,  $\text{I}^\ominus$
- Which of the following is correct regarding formation of reactive intermediates formed on heterolytic bond cleavage:  
 (1)  $\text{CH}_3 - \text{S} - \text{CH}_3 \longrightarrow \text{CH}_3^\ominus + \text{CH}_3 - \text{S}^\oplus$   
 (2)  $\text{CH}_3 - \text{Cu} \longrightarrow \text{CH}_3^\ominus + \text{Cu}^\oplus$   
 (3)  $\text{CH}_3 - \text{Cl} \longrightarrow \text{CH}_3^\ominus + \text{Cl}^\oplus$   
 (4) 1 & 2 both
- In which of the following labelled atom represents correct electrophilic centre:  
 (1)  $\text{CH}_3 - \text{CH} = \overset{*}{\text{O}}$       (2)  $\text{CH}_3 - \text{C} \equiv \overset{*}{\text{N}}$   
 (3)  $\overset{*}{\text{C}}\text{H}_3 - \text{I}$       (4)  $\text{CH}_3 - \text{CH} = \overset{*}{\text{N}}\text{H}$
- In nitromethane  $\text{CH}_3 - \text{N} \begin{matrix} \nearrow \text{O} \\ \searrow \text{O} \end{matrix}$ ,  
 two N – O bond lengths are:  
 (1)  $a = b$                                 (2)  $a > b$   
 (3)  $b > a$                                 (4) Can not be compared
- Which of the following is true:  
 (1) The molecule having more no. of equivalent canonical structures always has more resonance energy  
 (2) Actual structure of a molecule has energy lower than that of any of resonating structures.  
 (3) Resonance energy is the difference of energies of most and the least stable canonical structures  
 (4) 1 & 2 Both
- Correct resonating structure of acetate ion is:  
 (1)  $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\parallel}}\text{C} - \overset{\ominus}{\text{O}}$       (2)  $\text{H}_2\text{C} = \overset{\text{O}}{\underset{\text{OH}}{\mid}}\text{C} - \overset{\ominus}{\text{O}}$   
 (3)  $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\mid}}\text{C} - \overset{\ominus}{\text{O}}$       (4) All
- Correct order of stability of the following resonating structure is  

$$\text{CH}_2 = \overset{\text{O}}{\parallel}\text{CH} - \text{C} - \text{H} \longleftrightarrow \text{CH}_2^\oplus - \text{CH} = \overset{\ominus}{\text{O}} - \text{C} - \text{H}$$

(a)
(b)

$$\longleftrightarrow \text{CH}_2^\ominus - \text{CH} = \overset{\oplus}{\text{O}} - \text{C} - \text{H}$$

(c)

 (1)  $a > b > c$                                 (2)  $a > c > b$   
 (3)  $b > a > c$                                 (4)  $c > b > a$

11. Which of the following shows +R/+M effect:

- (1)  $>\text{C}=\text{O}$
- (2)  $-\text{X}$
- (3)  $-\text{COOH}$
- (4)  $-\text{CONHR}$

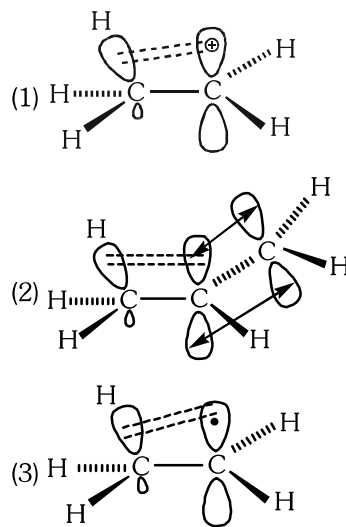
12. Which of the following compounds does not show electromeric effect

- (1)  $\text{CH}_3-\text{CN}$
- (2)  $\text{CH}_3-\text{CHO}$
- (3)  $\text{CH}_2=\text{CH}_2$
- (4)  $\text{CH}_3-\text{Cl}$

13. Which of the following is not an example of +E effect:

- (1) Hydrohalogenation of Propene
- (2) Hydroboration of propyne
- (3) Formation of acetaldehydecyanohydrine
- (4) Hydration of ethene

14. Hyperconjugation can be represented as:



(4) All

15. Hyperconjugation is not present in

- (1)  $\text{CH}_3-\ddot{\text{C}}\text{H}_2$
- (2)  $\text{CH}_3-\text{CH}=\text{CH}_2$
- (3)  $(\text{CH}_3)_3\text{C}-\text{CH}_2^+$
- (4) 1 & 3 both